The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

University College London Hospitals NHS Foundation Trust

DOUBLE PUMPING VASOACTIVE DRUGS in NEUROCRITICAL CARE

__Adrenaline__, __noradrenaline__, __dopamine__ and __dobutamine__ have very short biological half lives (1-2 minutes) due to reuptake in tissues - __these drugs must be double pumped at all times.__ There must be uninterrupted administration of these drugs, thus one infusion is commenced when the other is almost empty, this is termed 'double pumping' or 'piggybacking' – any interruption in flow will result in haemodynamic instability and can cause cardiac arrest. Drugs with a longer half-life do not require double pumping.

EXCLUSIONS

- Patients on ***high dose*** inotropes or with ***labile BP*** may require a more **patient specific approach**
- There is no literature defining a high dose of adrenaline and noradrenaline
- ICUs across UCLH have suggested high dose = **≥0.4mcg/kg/min**
- Patients on **dobutamine** may require a more **patient specific approach**
  a. It has vasodilatory properties as well as increasing HR and contractility
  b. As the dose increases these side effects can become more marked
  c. It needs to be double pumped but, while the majority of patients ***will not*** have the effects mentioned above, some will

STAFF CRITERIA

1. Registered Nurse accredited to administer intravenous medication as stated in UCLH Medicine Management, Prescribing, Administration and Dispensing of Medicines Policy 2022

and

1. Registered Nurse practising in the critical care environment who has been assessed as competent to perform the double pumping procedure

Nurses with less than 6 months experience in a critical care area will require supervision and to be observed performing the procedure a minimum of 3 times by a competent critical care practitioner

DOUBLE PUMPING GUIDELINE

If patient becomes profoundly hypotensive or hypertensive seek advice from a senior clinician (medical / nursing) immediately

INOTROPES SHOULD NOT BE GIVEN AS A BOLUS

STEP 1

Commence double pumping (syringe change) when there is:
EITHER
30 minutes to 1 hour remaining to run
OR
5mL of drug remaining in syringe *(whichever occurs first)*

STEP 2

Set new syringe to run at half infusion rate leaving current syringe rate unchanged

*In doses running **≤0.05mcg/kg/min** it may be necessary to set the new syringe to run at the full rate – i.e. both syringes run at same rate concurrently*

Immediately unclamp the second octopus lumen so both syringes are 'on' to the patient

Monitor arterial pressure closely - observe for an increase in systolic blood pressure (SBP)

The increase will be patient and inotrope dose dependent but **should not exceed 20mmHg**

Note: Extra care should be taken in patients with **subarachnoid haemorrhage** where prescribed SBP parameters should be taken into consideration

STEP 3

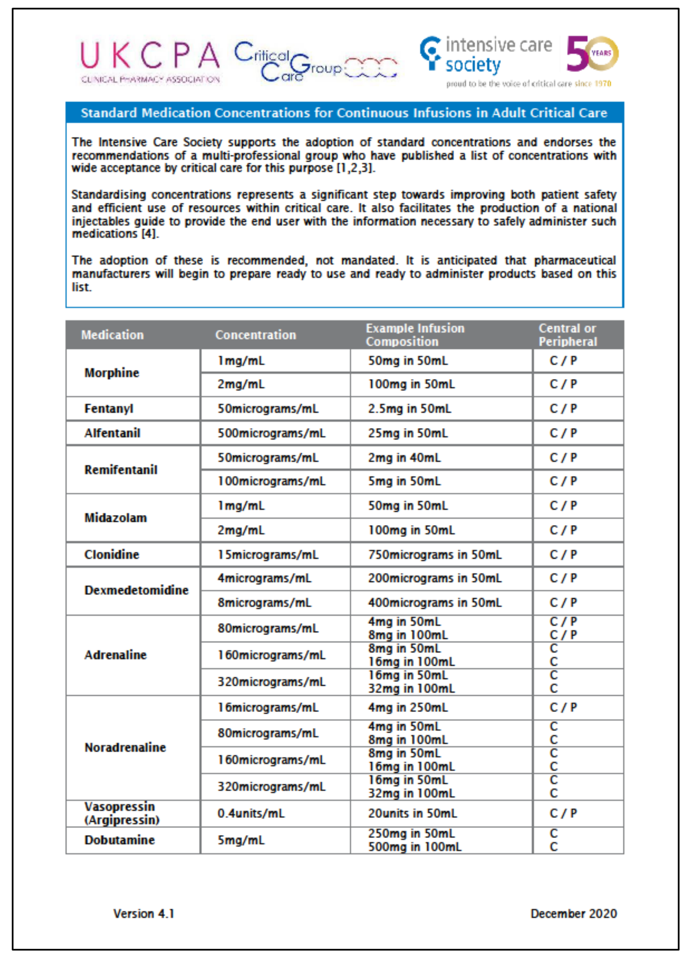
When **increase in SBP achieved** immediately increase new syringe to full rate, whilst decreasing original syringe to half original rate

Wean or turn old syringe off as indicated by SBP

When the patient's BP is established as stable following the double pumping procedure, clamp the octopus lumen to the original syringe then replace with a new syringe - this lumen should remain 'off' to the secondary syringe when not in use

LABELLING EXAMPLE

INOTROPE	SYRINGE DRIVER	PATIENT END
DRUG 1 – Noradrenaline	**1** Noradrenaline 8mg/50ml	**1** Noradrenaline 8mg/50ml Date
DRUG 2 – Noradrenaline	**2** Noradrenaline 8mg/50ml	**2** Noradrenaline 8mg/50ml Date
DRUG 3 – Adrenaline	**1** Adrenaline 4mg/50ml	**1** Adrenaline 4mg/50ml Date
DRUG 4 – Adrenaline	**2** Adrenaline 4mg/50ml	**2** Adrenaline 4mg/50ml Date



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